Non-LLM Routing: Three Training Recipes

Jonathan-aligned Config B outputs | HotpotQA: 4,814 train / 1,110 validation / 1,481 test

Qwen2.5-7B / 72B retrieval + Llama-3.2-3B / Llama-3.1-70B answering | Shared standard lenient accuracy

SS BASELINE **76.77% accuracy** 100.00% cost (929 Table 5 budget units) Direct strong + strong

Linear Softmax

TF-IDF + statistics → 2-way linear head

Training	Accuracy	Cost / SS
SFT	**76.50%**	107.78%
SFT + Reward	**76.64%**	107.94%
GRPO	**76.03%**	106.60%

MLP · 64 hidden units

TF-IDF + statistics → 64 ReLU → 2

Training	Accuracy	Cost / SS
SFT	**76.84%**	106.76%
SFT + Reward	**76.30%**	106.87%
GRPO	**76.70%**	108.21%

MLP · 128 → 64

TF-IDF + statistics → 128 → 64 → 2

Training	Accuracy	Cost / SS
SFT	**76.84%**	107.46%
SFT + Reward	**76.84%**	106.75%
GRPO	**76.77%**	108.61%

All rows: two stage routers · three-seed probability ensemble · thresholds selected for maximum validation accuracy

01 SFT

Learn W/S labels with weighted cross-entropy.

Retrieval labels: coverage in weak evidence.

QA labels: keep a correct weak answer;

otherwise upgrade if the strong answer works.

Unrecoverable cases map to W.

3 epochs · lr 1e−4 · checkpoint by val AUC

02 SFT + Reward

Start from SFT; evaluate both cached actions.

R = token F1 − 0.1 × C_train / 929

Optimize exact expected R + 0.03 × entropy.

Retrieval uses the best downstream QA reward.

Both stage routers are refined.

3 epochs · lr 1e−5 · checkpoint by val reward

03 GRPO

Start from SFT; sample 8 actions per row.

Normalize rewards within each sampled group.

Policy-gradient update + 0.02 reference KL.

Frozen SFT reference; no PPO clipping.

Stage adaptation of Jonathan’s public code.

2 epochs · lr 1e−4 · checkpoint by val reward

Cost: Table 5 prices, excluding decision-model compute. Every executed weak call is charged; fully escalated SS costs 1009 / 929 = 108.61%.

Reward uses the author’s training prices [80, 195, 894, 929]; evaluation uses [80, 195, 894, 1009]. GRPO is an adaptation, not a verified reproduction of the slide’s GRPO row.

Same frozen executor outputs and split across models. Previously inspected test: descriptive results. Models use train-fitted features; no LLM encoder or generation.